

**Impact of cleanser pH on maintaining a healthy skin barrier**

Stacy Hawkins, PhD, Unilever R&D; K. P. Ananthapadmanabhan, PhD, Unilever R&D



Maintaining skin surface pH in a mildly acidic range is important for maintaining a healthy stratum corneum barrier. The destabilizing effect of a high pH soap on lipid structure in skin compared to neutral pH synthetic detergent cleansers (syndets) is well known. This would suggest that lowering pH is a direct route to a milder cleanser; however, it is not known if all cleanser types with low pH are de facto milder. The objectives of this research was to understand the impact of lowering pH to skin barrier properties for both bar and body wash (regular anionic) cleansers. Healthy male and female subjects, ages 18-65 years old, provided informed consent to participate in randomized, double-blind, IRB-approved controlled wash studies to evaluate cleanser mildness (assessed by visual dryness and erythema and trans-epidermal water loss), and changes to skin surface pH following washing (both immediately and after up to two weeks of washing). Both bar and body wash products were evaluated. The pH of bar formulations was 5.5 compared to 7.0 using commercially available bars, and pH of 4, 7, and 10, respectively for the liquid body washes with the same base of a commonly used anionic surfactant (sodium laureth sulfate, or SLES). For the bars, the neutral pH bar was significantly milder than the bar with pH 5.5. Similarly, the pH-neutral body wash was significantly milder than the body washes with pH 4 and 10, and showed minimal change to surface pH immediately following washing. Keratin charge increases with decreased pH in anionic body washes, thereby increasing anionic surfactant binding. This potentially explains the mechanism for the reduced mildness with the low pH anionic surfactants evaluated here. Based on these results, the neutral pH systems were milder for both bar and body wash (traditional anionic) products evaluated. The results highlight the importance of pH in combination with the surfactant used for optimizing cleanser mildness.

*Commercial support: 100% is sponsored by Unilever R&D.*

**Impact of itch on health-related quality of life and work productivity and activity impairment: Results from a large observational physician and psoriasis patient survey**

Matthias Augustin, MD, University Medical Center Hamburg; Seong Jin Jo, MD, Seoul National University Hospital; Christopher Griffiths, MD, Centre for Dermatology; Ricardo Romiti, MD, Hospital das Clínicas University of São Paulo (USP); Esther Guevara-Sangines, MD, Hospital Regional "Lic. Adolfo López Mateos" ISSSTE México City; Tanya Howe, BS, GfK; Craig Richardson, BS, Novartis Pharma AG; Haijun Tian, PhD, Novartis Pharmaceuticals Corporation; Luigi Naldi, MD, Azienda Ospedaliera papa Giovanni XXIII

**Introduction:** Psoriasis is a chronic and highly symptomatic disease with itch as one of the most common and bothersome symptoms. This study explores the impact of itch on patients' health-related quality of life (HRQoL) and work productivity and activity impairment from an international survey of physicians and their systemic eligible psoriasis patients.

**Methods:** A cross-sectional survey of dermatologists and their systemic eligible psoriasis patients was conducted in EU5, Russia, Mexico, Brazil and South Korea between September 2015 and January 2016. Eligible patients currently have or have ever had moderate to severe psoriasis, and must have received a prescription treatment. Extent of itch was defined on a scale from 1 (no itch) to 10 (worst imaginable itch), and its severity was categorized as mild (1-3), moderate (4-7) and severe (8-10). Bivariate and multivariate analyses were conducted to assess the impact of itch vs. no itch and severity of itch on HRQoL (12-Item Short Form Health Survey [SF-12], Dermatology Life Quality Index [DLQI], EuroQol five dimensions [EQ-5D] questionnaire), sleep and work productivity and activity impairment (Work Productivity and Activity Impairment [WPAI] questionnaire).

**Results:** Of 3821 psoriasis patients, 1550 (41%) had itch at the time of the survey among whom 1010 (65%) and 305 (20%) had moderate and severe itch, respectively. Compared to patients without itch, patients with itch had a greater likelihood of skin pain (odds ratio (OR) = 1.8, confidence interval (CI) = 1.8 [1.4; 2.2]), and a significantly worse HRQoL (-1.4, -1.5, 1.3 and -0.03 change of scores for SF-12 Physical, SF-12 Mental, DLQI and EQ-5D, respectively, all  $P < .01$ ). Patients with itch also experienced a 16.9% (CI = [1.7%; 34.3%]) increase in work productivity impairment. Furthermore, greater severity of itch was associated with a greater likelihood of sleep disturbance (moderate itch vs. mild itch: OR = 1.9, CI = [1.3; 3.0]; severe itch vs. mild itch: OR = 3.5, CI = [2.1; 5.7]). Compared to mild itch, both moderate and severe itch were also associated with worse HRQoL and greater work productivity impairment ( $P < .05$ ).

**Conclusion:** The results reveal a consistently greater burden associated with itch and severity of itch amongst psoriasis patients. These results confirm that reducing itch in psoriasis is an important patient-relevant treatment goal that should be monitored in a more systematic manner in clinical practice.

*Commercial support: This study is syndicated in nature and the data is owned by GfK. This abstract was developed for and funded by Novartis Pharma AG, Basel, Switzerland.*

**Impact of cutaneous toxicities in patients taking vandetanib**

Megan Rogge, MD, Icahn School of Medicine at Mount Sinai Hospital; Anisha Patel, MD, UT MD Anderson Cancer Center



**Background:** Vandetanib is an oral multikinase inhibitor that targets the RET (rearranged during transfection) kinase, VEGFR (vascular endothelial growth factor receptor), and EGFR (epidermal growth factor receptor). Previous clinical trials investigating vandetanib have reported cutaneous toxicities, however, we sought to determine the incidence and time to onset of cutaneous toxicities in patients enrolled in clinical trials at our center investigating vandetanib as a single agent or in combination. We also aimed to describe the impact of cutaneous toxicities on treatment course and tumor response.

**Methods:** We used natural language processing to search the "transcribed documents" portion of our electronic medical record for patients who were receiving vandetanib and had a cutaneous toxicity between January 2009 and May 2016. Patients' charts were reviewed to confirm presence of cutaneous toxicity and time to onset. We also noted if the vandetanib dose was altered due to cutaneous toxicity. Tumor outcomes were assessed using RECIST criteria.

**Results:** We identified 55 patients with cutaneous toxicities on vandetanib monotherapy or combination treatment. Among the 17 patients receiving vandetanib monotherapy, photo-induced drug rash was the most common (13/17 patients) with a median latency of 9.5 weeks. In those with everolimus co-therapy (34 patients), the most common cutaneous toxicities were mucositis (22/34 patients) and acneiform eruption (18/34 patients), both with a median latency of 2 weeks. Less commonly, a photo-induced drug rash (9/34 patients), Hand Foot Syndrome (HFS) (3/34 patients), panniculitis (3/34 patients), and an uncharacterized rash (9/34 patients) were described. Seventeen of the total 55 patients with a cutaneous toxicity (31%) had either a dose decrease, interruption or discontinuation of vandetanib due to the cutaneous toxicity. Among all patients taking the combination of vandetanib and everolimus (95 patients), 11% had a decrease in dose, interruption, or discontinuation. There was no association between cutaneous toxicity and tumor response.

**Conclusion:** Cutaneous toxicities in patients taking vandetanib include photosensitivity, acneiform eruption, and mucositis and are a significant barrier to a sustained course of treatment. Patient education and early dermatology involvement may help decrease the impact on cancer therapy. Unlike the acneiform eruption in EGFR inhibitors, vandetanib-induced acneiform eruption does not appear to correlate with tumor response.

*Commercial support: None identified.*